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Section : BSCS 3B
Subject : CSST102 – Basic Machine Learning
Assessment : Assessment Task No. 1 - Introduction to Machine Learning

From Data to Prediction: Understanding Machine Learning-Based Product Recommendation

Chosen ML Application:

Product Recommendation - a system that suggests products to a customer while they are shopping on an online store.

Problem Description:

It's common for online shopping platforms to list thousands or even millions of products, making it difficult to look for interesting products. To overcome this problem, a product recommendation system uses the customer's previous activities, such as the products they have viewed, purchased, added to cart, or clicked, to be more helpful in the selection process. From these activities, the system can identify patterns and recommend other products that the customer might want. This can be useful for both the customer and the business. The customer can find products more easily, while the business can potentially increase sales because the recommendations are more personalized instead of showing the same products to everyone.

Input and Output Identification:

The table below identifies the application, problem, inputs, outputs, patterns, and data needed, following the same format used in the lesson's video recommendation example.

Question	Your Answer
1. What is the application?	Product recommendation for an online shopping platform. For example, e-commerce app like Shopee or Lazada.
2. What problem are you trying to solve?	Recommend products to a customer based on their past browsing and purchase behavior.
3. What are the inputs?	Customer's purchase history, viewed items, added items to cart, and product category preferences.
4. What is the expected output?	A list of recommended products that match the customer's interest.
5. What patterns or relationships might the ML system learn?	Customers who buy or view items from a certain category that tend to buy more items from the same or related category.

6. What data would the system need?	Product IDs, product categories, purchase timestamps, browsing history, and customer ratings or clicks.
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Traditional Programming vs. Machine Learning:

In traditional programming, the developers need to know what the rules are. The flow of traditional programming is Input with an Algorithm then it produces an Output. For a System that has a product recommendation feature, a programmer using this kind of approach have write the rules manually, like "if the customer bought bicycle, recommend a bicycle parts," for every single product combination. This will be very hard to maintain, because there are thousands of products and the rules need a constant update every time new products or new customer behavior appears.

In Machine Learning, the flow is different, you need an Input and Output or Data then Machine Learning Algorithm and the model learned the relationship. Instead of the programmer writing the rules, the algorithm is given many data of customer behavior and the products they eventually bought, and it will learn the pattern by itself. This is more scalable, since the model can automatically adjust its recommendations when there is a new products, new customers, new trends comes in, without having to rewrite the rules manually.

Mini Dataset:

Below is a simple dataset with 5 observations showing the relationship between how many items a customer bought from a certain category and how many similar items the system recommends afterward.

Input (No. of items bought in the same category)	Output (No. of similar products recommended)
1 (ex. Phone Case)	2 (Screen Protector, Ring Holder)
2	4
3	6
4	8
5	?

Pattern/Relationship:

So basically, what I notice in the pattern is every time a customer buys one item in a category the system will recommend double number of similar products to them. Like if they buy 1 item then 2 products will show up, if they buy 2 items then 4 products will show up and so on. So, the missing number when input is 5, the output must be 10 because it just keeps doubling every time.

If we going to write it in math way it looks like this, **Output = 2 x Input or $f(x) = 2x$** that means whatever number you put in the input side you just multiply it by 2 to get the output.

Now for how a machine learning could learn this same pattern, basically the computer will look at all the input and output that we already have and it will try different guesses or rules to see which one match the best. It keeps comparing its guess to the real answer and check how far or how close it is, then it adjusts itself again and again until the mistake becomes small or almost zero. After doing that many times it finally learns that the rule is always double the input, so now even if we give it a new number that it never seen before like 5, it can already predict that the answer should be 10 because it already learns the pattern from before.

Explanation of the Training Process:

How does a Machine Learning algorithm learn from data?

So basically, how it works is, the algorithm first gets some input, like things the customer already bought before, and then we also show it the right answer too, which is the things the customer actually bought or liked after. And while its learning, the model just keeps trying and trying to match the input with the output, and every time it makes a guess it will check how close or how far that guess is from the real answer.

And every time it guesses wrong, it changes itself a little bit so next time the mistake is smaller. It keeps doing this again and again with lots of data, so slowly it will start to learn the real connection between the input and the output. And after it done learning all that, now it can guess on new data that it never even seen before, and that is basically how it makes new product recommendation for the customers.

Short Reflection:

Why is learning from data useful instead of manually programming every possible rule?

Learning with data is more useful than manual programming because it is almost impossible for a programmer to manually make rules, especially for stores that have thousands or millions of customers and products, and there are many stores also in an e-commerce system, so it is tedious to do that. Learning with data is also useful for other complex tasks like language translation and prediction, which have many rules and exceptions to even code by hand. So I use a Machine Learning based approach for the product recommendation system, since it involves patterns that are too complex for a human to define manually. So, letting the machine learning algorithm discover the pattern by learning from real customer data is much more practical and scalable than using traditional programming.